
A
Per-module ROC grid: signal decomposition across outcomes
Each cell = univariate AUC from one module eigengene.
Border: solid black = BH q<0.05; dashed = nominal p<0.05; grey = ns. 1000 permutations (Phipson–Smyth adjusted); paired within subject for Pre vs Post rows.
Age (Y vs O)
Combined ME
Δ VL responder
Pre ME
Pre vs Post (all)
Raw ME
Pre vs Post (Y)
Raw ME
Pre vs Post (O)
Raw ME
Baseline LBM
Pre ME
Turquoise
AUC 0.96 [0.91, 1.00] p < .001
AUC 0.60 [0.38, 0.82] p = 0.345
AUC 0.52 [0.38, 0.67] p = 0.649
AUC 0.56 [0.34, 0.78] p = 0.552
AUC 0.56 [0.34, 0.78] p = 0.598
AUC 0.58 [0.37, 0.79] p = 0.421
Blue
AUC 0.90 [0.78, 1.00] p < .001
AUC 0.66 [0.45, 0.86] p = 0.141
AUC 0.73 [0.60, 0.85] p < .001
AUC 0.86 [0.71, 1.00] p < .001
AUC 0.71 [0.51, 0.90] p = 0.065
AUC 0.57 [0.36, 0.79] p = 0.468
Brown
AUC 0.60 [0.38, 0.81] p = 0.377
AUC 0.59 [0.38, 0.81] p = 0.410
AUC 0.52 [0.37, 0.67] p = 0.737
AUC 0.53 [0.32, 0.75] p = 0.650
AUC 0.56 [0.35, 0.78] p = 0.499
AUC 0.46 [0.24, 0.68] p = 0.683
Yellow
AUC 0.80 [0.62, 0.97] p = 0.004
AUC 0.49 [0.27, 0.71] p = 0.929
AUC 0.60 [0.45, 0.75] p = 0.066
AUC 0.72 [0.53, 0.91] p = 0.024
AUC 0.55 [0.33, 0.77] p = 0.576
AUC 0.55 [0.33, 0.77] p = 0.671
Green
AUC 0.74 [0.55, 0.92] p = 0.022
AUC 0.72 [0.52, 0.91] p = 0.050
AUC 0.58 [0.43, 0.73] p = 0.196
AUC 0.73 [0.54, 0.93] p = 0.015
AUC 0.48 [0.26, 0.70] p = 0.860
AUC 0.60 [0.39, 0.81] p = 0.360
Red
AUC 0.72 [0.53, 0.91] p = 0.037
AUC 0.48 [0.26, 0.70] p = 0.823
AUC 0.57 [0.42, 0.72] p = 0.295
AUC 0.55 [0.32, 0.77] p = 0.657
AUC 0.61 [0.41, 0.82] p = 0.253
AUC 0.46 [0.24, 0.67] p = 0.713
Black
AUC 1.00 [1.00, 1.00] p < .001
AUC 0.58 [0.37, 0.80] p = 0.455
AUC 0.47 [0.32, 0.62] p = 0.550
AUC 0.67 [0.47, 0.87] p = 0.124
AUC 0.48 [0.26, 0.70] p = 0.843
AUC 0.58 [0.36, 0.80] p = 0.485
Pink
AUC 0.56 [0.34, 0.77] p = 0.618
AUC 0.63 [0.42, 0.84] p = 0.227
AUC 0.58 [0.44, 0.73] p = 0.270
AUC 0.63 [0.42, 0.84] p = 0.250
AUC 0.55 [0.33, 0.77] p = 0.639
AUC 0.60 [0.38, 0.81] p = 0.373
Row 1 (Age): Combined ME = (Pre+Post)/2 per subject, n=30. Row 2 (ΔVL responder): median-split of age-residualized ΔVL, predictor = Pre ME. Rows 4-5 (age-specific Pre vs Post): paired within subject; Young n=15 × 2, Old n=15 × 2. Row 6 (LBM): median-split of age-residualized baseline LBM, predictor = Pre ME. Row 3 (Pre vs Post): raw ME values stacked across 30 subjects.
B
Full sweep: module eigengenes vs Δphenotype, age-stratified
Each cell = per-age Pearson r (Young solid / Old dashed OLS).
Border: solid black = BH q<0.05; dashed = nominal p<0.05 in Young; grey = ns. Screen: 180 tests; 2 raw-sig, 0 BH-sig.
Predictor: Δ ME
Predictor: Baseline ME
Turquoise
Blue
Brown
Yellow
Green
Red
Black
Pink
Magenta
ΔVL
ΔLBM
ΔDL
ΔT1
ΔT2
Inline stats = Pearson r per stratum (Y, O). Figure 6C shows the six cells with the lowest raw p: Δ Odd. Phos. | Complex I vs ΔVL, Δ Odd. Phos. | Complex I vs ΔT2, baseline Muscle Contraction | Sarcomere vs ΔT2, baseline Protein Degrad. | Proteasome vs ΔVL, baseline Lipid Catabolism | TCA Cycle vs ΔVL, baseline Protein Folding | Chaperones vs ΔLBM.